Operating Systems and Networks

Computing and Mathematics

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Operating Systems and Networks (A14065)

Short Title:	Operating Systems and Networks		
Faculty	Science and Computing		
Department	Computing and Mathematics		
Cluster	Networks and Cloud		
Author	DOHALLORAN		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

This module introduces the fundamental concepts of operating systems and computer networking. The module will concentrate on core operating system components such as memory management, process management, file systems and virtualisation. The module will provide the student with an understanding of networking terminology, protocol models and devices. A brief introduction to routing and wireless technologies will also be provided. The student will use simulation and virtualisation tools in the practical element of this module.

Programmes

COMP-0532	Diploma in Computing with Security and Forensics (WD_BCSEC_SP)	3 / 2 / M

Indicative Content

- Operating system structure, components, services and utilities, virtualisation
- Process Management: in-memory process data structures, concurrency, threads, CPU scheduling, interprocess communication
- Memory management: memory hierarchy
- Virtual memory
- File management
- Networking concepts: OSI Model, TCP/IP overview; LAN Technologies (Ethernet), IP addressing
- Routing: static routes and various dynamic routing protocols between LANs
- Wireless Technologies: 802 Architecture

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Investigate the memory management, process management and file management components of a modern operating system with regard to operation principles, data structure requirements and algorithms used.
2. Discuss the basic concepts and components of networking.
3. Configure a contemporary operating system through the use of command line instructions and scripts.
4. Communicate the main concepts of wireless and mobile networks.
5. Investigate the IP routing process and design and implement an IP addressing scheme for a given network scenario.

Learning and Teaching Methods

- This module will be presented by a combination of lectures and practicals.
- This module adopts a hands-on practical approach with weekly laboratories that make use of virtualisation and simulation tools.

- Self-directed learning.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,5
Continuous Assessment	50%	
Practical	30%	3,4,5
In-Class Assessment	20%	3,4,5

Assessment Criteria

<40%: Unable to interpret and describe key concepts of operating systems and networks.

40%-49%: Be able to interpret and describe key concepts of operating systems and networks.

50%-59%: Ability to discuss key concepts of operating systems and networks. Be able to perform basic configuration and administration tasks.

60%-69%: All of the above, in addition, be able to solve problems within the specific knowledge domain(s) by experimenting with the appropriate skills and tools.

70%-100%: All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of both complex and unforeseen problems through the use of appropriate skills and tools.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Supplementary Material

- Kurose, J. and K. Ross. _Computer Networking: A Top-Down Approach_. 6th Ed. Boston: Pearson Education, 2012.
- Silberschatz, A., P. Galvin and G. Gagne. _Operating Systems Concepts_. 9th Ed. New York: Wiley, 2013.
- Stallings, W. _Operating Systems: Internals and Design Principals_. 8th Ed. New York: Pearson Education, 2014.

Requested Resources

- Room Type: Computer Lab